Distributed EV charging network

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Technical report and verified study record - 22 October 2026

Scope and evidence

The existing C simulator is preserved with two targeted fixes: the discovery timestamp is captured when congestion is detected, before the global alert gather can delay reporting; and enormous finite tick arguments are rejected before conversion to the system time type. Its strict build, 25 local integration runs, expanded independent candidate oracle and sanitizer checks passed during this audit. The supplied 540-run local TCP-container study was independently recalculated from all 107,208 retained raw records. A separate CAAS study now provides measurements from two allocated compute nodes; local and CAAS results remain explicitly separate.

CAAS job 199603 completed successfully. The retained evidence covers 119 simulations: 24 correctness runs, 90 scaling runs, four expiry runs and one run with three ten-second update cycles. There are five grid sizes and five line sizes, three implementations and three launches per scaling case. The full 30-grid/30-line requirement is supplied by the local study, as permitted by caas_instructions/caas_usage.txt.

Architecture and port simulation

World rank 0 is the only base station. MPI_Comm_split excludes it from a nonperiodic MPI_Cart_create worker grid. Cartesian rank i is world rank i+1; reordering is disabled. MPI_Cart_shift supplies the four orthogonal neighbours, with MPI_PROC_NULL at missing edges. Exactly rows*cols+1 MPI ranks are required.

Each station creates one persistent OpenMP thread per port. The shared ports[k] array stores 1 for occupied and 0 for free. Each thread owns one port element and private xorshift state. An atomic sum counts free ports; barriers publish each complete snapshot. Only the initial thread calls MPI, under MPI_THREAD_FUNNELED. The base creates no OpenMP port team.

Default simulation choices are threshold 1, tick 10 seconds, expiry 30 seconds, ten rounds and seed 3143. The user must supply rows, columns and ports. Random mode independently chooses an occupied state with probability 0.75 per port per update; this is a synthetic model. Full, empty, ties and mixed scenarios provide deterministic checks.

AI declaration

OpenAI Codex substantially assisted the original implementation and documentation and this audit, including code review, additional validation, CAAS configuration and execution, results analysis, graphs, slides and packaging. This does not claim independent student authorship of generated material. Both students must understand and verify the submitted work. The available original and current user prompt records are preserved in AI_Prompt_Records.pdf; unprovided histories have not been reconstructed. No assessed presentation, interview or Moodle submission was performed by the assistant.

Protocol and implementation variants

One simulation cycle

1. The base broadcasts CONTINUE. Port threads update the shared snapshot and count availability.
2. Every station exchanges one integer query intent per real neighbour. Zero intent is still sent, so all peers execute a matching deterministic phase. Replies carry one integer capacity only where requested.
3. A station below its availability threshold selects a suitable immediate neighbour, preferring the lowest rank. If no neighbour qualifies, it sends one packed report to the base after an alert-flag gather.
4. The base receives the complete report batch and updates every cache entry before computing advice. It replies with a count and all equally near candidate ranks. The station log represents forwarding that advice to simulated vehicles.
5. Stations wait for the configured tick and complete a global round barrier.

Reports contain station rank, coordinates, cycle, UTC discovery time, local and neighbour capacities, and neighbour message counts and payload bytes. An alert has no suitable immediate neighbour; query intents and replies sent and received by that station are counted in its report.

Base-station information and all ties

The search follows Task 2's example: source 0 in a 3x3 grid has the distance-two frontier {2,4,6}. Manhattan distance matches orthogonal adjacency. The base excludes candidates with a recent self-alert or fresh observed capacity below threshold. Missing or expired evidence permits an inferred candidate; it does not prove a vacancy. All surviving ties are returned in rank order, or an empty list. The search does not silently extend beyond this frontier.

V	Neighbour phase	Base receives / replies	Candidate search
V0	Blocking MPI_Sendrecv	Rank-ordered MPI_Recv / MPI_Send	Scan N nodes
V1	MPI_Irecv / MPI_Isend / MPI_Waitall	MPI_Probe ANY_SOURCE / MPI_Send	At most 8 offsets
V2	MPI_Irecv / MPI_Isend / MPI_Waitall	Preposted MPI_Irecv / batched MPI_Isend	At most 8 offsets

Computational and communication costs

With N stations, k ports, A alerts and station degree d<=4, a port snapshot takes O(k) total work and O(k) storage, with O(1) work per port thread plus synchronization. The grid has E=m(n-1)+n(m-1) undirected edges. Each round sends 2E query intents and at most 2E replies, plus 2A report/advice messages. Shutdown sends N acknowledgements; MPI internal collective packets are not counted as application messages.

V0 candidate work is O(N) per alert and O(A*N) per cycle. V1/V2 inspect at most eight frontier offsets: O(1) per alert and O(A) per cycle. All ties fit in eight ranks (two in a line). Base caches and report/request batches use O(N) storage. Persistent port teams avoid repeated thread creation. Nonblocking calls permit concurrency; the implementation does not claim independent computation/communication overlap.

Measurement method

Every interval starts and stops with MPI_Wtime on the same rank. No timestamp subtraction crosses hosts. Report completion may reflect eager buffering; redirect wait and round-trip time include base processing and queueing. These are endpoint operation costs, not pure wire latency. No station timer directly wraps sleep, startup or communicator setup. The plotted station barrier starts after its own tick, but includes imbalance; the retained base barrier can include waiting for station ticks. Simulation elapsed time ends after STOP and all ACKs; launch wall time also includes startup and finalization. The final instrumentation gather is outside the simulated WSN metrics.

Metric	Measured interval	Station normalization
query / reply	Neighbour phase elapsed time	Measured rounds
report	Station MPI_Send completion	Station report sends
redirect	Station receive wait, including base queueing	Advice receives
control	CONTINUE and final STOP broadcast	Measured broadcasts (rounds+1)
flags	Alert gather elapsed time	Measured rounds
sync	Global round barrier elapsed time	Measured rounds
ack	Station shutdown send completion	One per station per launch
alert_roundtrip	Report send through advice receipt	Alerts in measured rounds

Raw files retain totals for every rank, all nine classes, logical outgoing calls and application payload bytes. Query/reply calls count outgoing messages, excluding MPI_PROC_NULL; they do not count every Irecv/Waitall API invocation. Phase durations are divided by rounds. Each run mean gives equal weight to active station ranks after this normalization. Base-rank totals remain in the raw CSV for independent inspection. Collective buffer bytes exclude MPI protocol overhead.

Experimental controls

Study	Coverage and workload	Resource / timing limits
Supplied local study	30 grid sizes + 30 lines; V0/V1/V2; 3 launches; 2 warmup + 6 measured rounds; 2 ports; mixed congestion	Two TCP containers on one 16-CPU WSL host; explicit oversubscription; no artificial link delay
CAAS study	Grids 2x2, 2x3, 3x3, 3x4, 3x5; lines 4,6,9,12,15; all 3 variants x 3 launches; 2 warmup + 12 measured rounds	Two allocated compute nodes; 16 rank slots x 2 cores; Open MPI 4.1.5 / GCC 11.2.0; bound cores; passive OpenMP waits

Version order rotates deterministically between repeats. Plot points are medians of three launch means, with observed min-max ranges. ACK is one observation per station per launch. Local and CAAS protocols differ in measured cycle count, placement and host sharing, so their absolute delays are not presented as a controlled hardware speedup experiment. scripts/analyze.py and scripts/analyze_caas.py retain and check the aggregation rules.

Local study: measured results

These are the supplied 18 October 2026 container measurements, independently checked during this audit. All 540 scaling cases have both container hostnames and complete station shutdown acknowledgements. All 4,860 run means, 1,620 plotted summaries, comparison ratios and variant-selection scores recompute from the raw CSV.

Topology / metric	Median V0/V1	Median V0/V2
grid / query	2.581	2.272
grid / report	0.938	0.898
grid / redirect	1.086	0.987
grid / alert_roundtrip	1.095	0.986
line / query	3.024	2.762
line / report	0.982	1.022
line / redirect	0.844	1.042
line / alert_roundtrip	0.850	1.042

Ratios above one favour the refined implementation. The median alert-response criterion across all 60 matched sizes scores V0=1.000, V1=0.987, V2=1.018. This is a marginal descriptive preference for V2 on that shared host, not evidence of statistical significance or a universal best implementation.

Task 3 answers: grid and line scaling

Metric	Grid: first / last 5 (us)	Grid ratio	Line: first / last 5 (us)	Line ratio
query	3312.21 / 52526.58	15.86	27.72 / 28466.81	1026.80
reply	1878.64 / 28205.51	15.01	19.88 / 18665.77	938.95
report	9.05 / 20.66	2.28	5.43 / 26.21	4.82
redirect	4336.81 / 79396.58	18.31	255.48 / 90396.17	353.82
control	557.77 / 48136.42	86.30	38.45 / 26639.90	692.83
flags	678.06 / 14698.84	21.68	11.46 / 12613.40	1100.52
sync	4314.61 / 108662.00	25.18	118.54 / 94030.18	793.23
ack	43.41 / 62.32	1.44	18.20 / 89.93	4.94
alert_roundtrip	4346.91 / 79424.81	18.27	260.26 / 90419.46	347.42

For V2, increasing grid size raises observed neighbour, collective and response waiting costs; scheduling, synchronization and base fan-in contribute to these curves. Each station still has at most four neighbours, so ideal local communication work remains bounded while total traffic grows O(N). The irregular observations do not justify a universal linear measured-latency law.

A line reduces degree to at most two while retaining global collectives and one base. Its diameter is N-1, but a neighbour exchange uses one logical edge and base messages are direct MPI channels. Diameter therefore does not make one such message require N forwarding hops. The observed line curves also include oversubscription and shared-host scheduling.

CAAS: verified experimental results

Successful job 199603 ran on student-caas-n10, student-caas-n11. SLURM accounting records COMPLETED with exit code 0:0. Every study simulation returned normally, recorded all station ACKs and produced the expected per-rank metrics. The complete command log, placement map, base and station logs, scheduler output and source hash are retained under results/caas/.

Sixty launches with uneven task counts emitted a Slurm warning that the per-node cap was ignored. Independently checked placement remains balanced across both hosts and never exceeds eight ranks per node, with two requested cores per rank. All command and job-step exits are zero; no unexpected stderr or resource error is accepted by the validator.

Metric	Grid V0/V1	Grid V0/V2	Line V0/V1	Line V0/V2
query	1.639	1.806	2.341	2.492
reply	1.720	1.769	1.220	1.250
report	0.921	0.897	0.964	1.061
redirect	1.148	1.093	1.246	1.302
control	0.819	0.932	0.889	1.095
flags	1.184	1.460	1.440	1.476
sync	1.029	1.039	1.148	1.110
ack	0.825	0.978	0.978	1.010
alert_roundtrip	1.148	1.093	1.246	1.302

Across ten matched sizes, the alert-response selection scores are V0=1.000, V1=1.157, V2=1.197. The highest observed score is V2. This comparison is descriptive. Three launches and a small representative CAAS size set do not establish a universal optimum. The local 60-size comparison and the CAAS comparison are kept separate.

Expiry sensitivity

Expiry (seconds)	Alerts	Candidate recommendations
0	24	6
0.1	24	6
1	24	0
30	24	0

Expiry tests use six rounds and an accelerated 0.05-second tick. Short expiry can forget earlier congestion sooner; a long expiry can hide recovery. Inferred availability requires confirmation or reservation in a real EV service. The separate realistic sample uses three ten-second ticks and is not mixed into accelerated scaling data.

CAAS: benchmark and recovery

Isolated candidate benchmark

results/caas/candidate_benchmark.csv records 10,000 actual candidate calls at N=10,100,1000,10000 for each version. All checksums match the independent expected values. These times isolate candidate computation and exclude MPI exchange and port-thread costs; they must not be quoted as whole-program speedups.

Nodes	V0 seconds	V1 seconds	V2 seconds	Checksum
10	0.000424762	0.000232499	0.00023799	16000
100	0.00409147	0.000240133	0.000229484	19600
1000	0.0394466	0.000234573	0.000229534	19960
10000	0.392704	0.000229474	0.000229663	19996

Failed attempts and verified recovery

Job 199444 compiled successfully, then timed out in the initial forced-PMIx srun unit-test launch before any simulation evidence. Job 199460 failed when an Open MPI remote daemon did not connect back to mpirun. Diagnostic job 199513 tested launch routes separately: srun modes none and pmi2 completed the final WSN program across two nodes with all ACKs; forced pmix and TCP-forced mpirun timed out. A preliminary five-minute batch, 199520, exceeded its reservation before completing the study. The final fresh job uses ten minutes and the demonstrated srun --mpi=none --exact route with explicit ranks, per-node limits and two bound cores per rank. Failed/incomplete attempts and diagnostic probes are labelled separately; their timings are excluded from the final 90-run scaling comparison. Source logs and accounting are retained for verification.

Correctness, provenance and limitations

Checks actually performed

The local strict build uses C11, OpenMP, -Wall -Wextra -Wpedantic -Werror. Twenty-five local integration runs cover isolated full/empty stations, horizontal and vertical lines, simultaneous full congestion, all ties, empty rectangular grids, random ports, nondefault thresholds and expiry, and invalid arguments/process counts. All three variants return identical deterministic outcomes.

The expanded C oracle independently enumerates every rank for every 1..7 by 1..7 shape under four cache states and all versions. It checks corners, all eight two-hop candidates, expiry boundaries and recent low observations. Address/undefined-behaviour sanitizers passed that oracle. CAAS repeats 24 scenario/version integration runs plus the candidate tests. Its consolidation checks source equality, successful exits, every metric row, message and byte totals, report fields, two compute hosts and shutdown acknowledgements.

Source-to-results traceability

Current src/wsn.c SHA-256:
`28cd1d3951e9b4fa523765811bff2f3da3c747916e08a66d0d4e665b562fdcdd`

The source hash matches the CAAS environment record and measured source snapshot. Relative to the supplied local timed_source_snapshot.c, the final source adds identification comments and a final aggregate summary after the measured intervals, corrects a flush/close comment, moves UTC discovery-time capture before the alert gather, and checks the tick conversion range during argument parsing. Candidate decisions and communication methods are unchanged, but the old local measurements are explicitly tied to their retained old source rather than claimed as measurements of byte-identical final code. CAAS runs measure the final source. New analysis scripts reproduce derived results without inventing or modifying raw measurements.

Normal shutdown

The base broadcasts STOP after the configured cycles. OpenMP teams observe it at a barrier and exit. All nonblocking request batches have completed before their buffers are released. Stations flush logs, send ACKs and free port/PRNG arrays. The base receives all N ACKs, closes its log, and the final metrics gather completes. Files close, communicators are freed and all ranks call MPI_Finalize. Normal runs use no Ctrl+C or kill. Invalid arguments, allocation failures and I/O failures use MPI_Abort as an explicit error path.

Practical limits

This is a phased simulator with global gather/barrier costs and one base-station fan-in. The workload is synthetic; there is no measured radio interference, packet loss, road geometry or real EV arrival trace. Silence means inferred capacity, not a guarantee. Reports use MPI_BYTE with a native C struct and therefore assume homogeneous layouts. Larger or heterogeneous deployments need portable serialization, more realistic workloads, candidate confirmation and a different progress/termination design.

CAAS proves execution across distinct allocated compute hosts; it does not reveal whether their underlying physical hardware is independent. The local containers share a host and oversubscribe it. Small samples and host scheduling limit performance generalization. These limits remain explicit in the slides and results.

Presentation and assessment readiness

Presentation plan

Keep the 13-slide main narrative within a rehearsed 6-7 minutes. The planned timing is approximately 6:45, including pauses and pointing. Main slides explain the architecture, port synchronization, base cache, three variants, local grid/line/control/response curves, CAAS evidence, expiry and limitations, then Q&A.; Detailed per-category charts and execution evidence are clearly labelled appendices. Rehearsal duration and live delivery are not verified by file checks.

Q&A; preparation - answer without tools

Why MPI_THREAD_FUNNELED? Only the initial thread calls MPI.
Why can Send finish early? Eager buffering can make the send buffer reusable before receiver consumption.
Why prepost receives? Matching is ready before sends and does not rely on eager buffering.
What protects the port snapshot? One writer per element, atomic aggregation and OpenMP barriers.
What does Waitall prove? Completion and safe buffer reuse, not computation/communication overlap.
Why all ties and expiry? The specification requires all equally near candidates and a configurable recent-report window.
Why is a line's diameter insufficient to explain message latency? The implementation does not forward each message across every station.
Why can V2 lose? Request setup, synchronization, scheduling and eager protocols may outweigh concurrency for small messages.

Primary assessment and execution sources

docs/Lab #3 - Assessment Specification.pdf, especially pages 5-8, defines Tasks 1-4 and the submission checklist. docs/Lab #3 Rubric.pdf, HD column on pages 1-5, adds multiple variants, 30 sizes per topology and presentation/Q&A; standards. caas_instructions/Student-CAAS for FIT3143.pdf and CAAS_DEMO.zip supply account/module/SLURM guidance. The newer caas_usage.txt permits a representative 4-5 CAAS grid set and specifies a 20-minute partition limit; the live partition was checked before submission. These paths refer to the project reference folders, not files the specification requires the team to re-upload.

Relevant teaching examples are context/WarmUpTask_1.c, WarmUpTask_2.c and MPI_VirtualTopologyExample.c. They support Cartesian communicators, communicator splitting and master/worker MPI. The submitted implementation uses its own safe matching and persistent OpenMP scheme.

Submission and remaining student actions

Use README.md and SUBMISSION_CHECKLIST.csv in submission_ready for exact paths and classifications. The specification requires all C/C++ sources, sample base logs, available CAAS scripts, all timing logs and available graphs, presentation/documentation, and PDF prompt records when AI was used. This report incorporates the AI declaration, so no separate declaration PDF is needed. Both students must upload the same files before the allocated class, attend and present, and answer without AI. Moodle upload restrictions, attendance, rehearsal and oral understanding cannot be verified from these files.